



Figure 3: The world of the GameShell handheld gaming console

Clockwork Pi development board

Clockwork Pi (CPI) is designed to be hackable, being powered by an Allwinner R16 processor to run Linux, as well as an Arduino compatible Atmel AVR MCU.

Its technical specifications are:

- Allwinner R16-J quad-core Cortex A7 CPU @1.2GHz with Mali-400 MP2 GPU
- 1GB DDR3 memory
- MicroSD card slot with support up to 128GB of memory
- Micro HDMI output (720p)
- Wi-Fi (802.11 b/g/n) and Bluetooth 4.0 on-board
- USB OTG supported
- Small size and with PMU (power management unit)
- Audio output (3.5mm headphone jack)

The Clockwork Pi operating system

The Clockwork Pi operating system comes preinstalled on an SD card and is based on Debian 9 ARMhf and Linux kernel 4.1x. It supports the running of

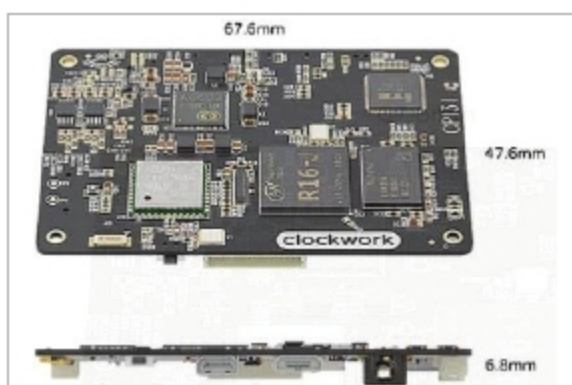


Figure 4: Clockwork Pi (CPI v3.1)

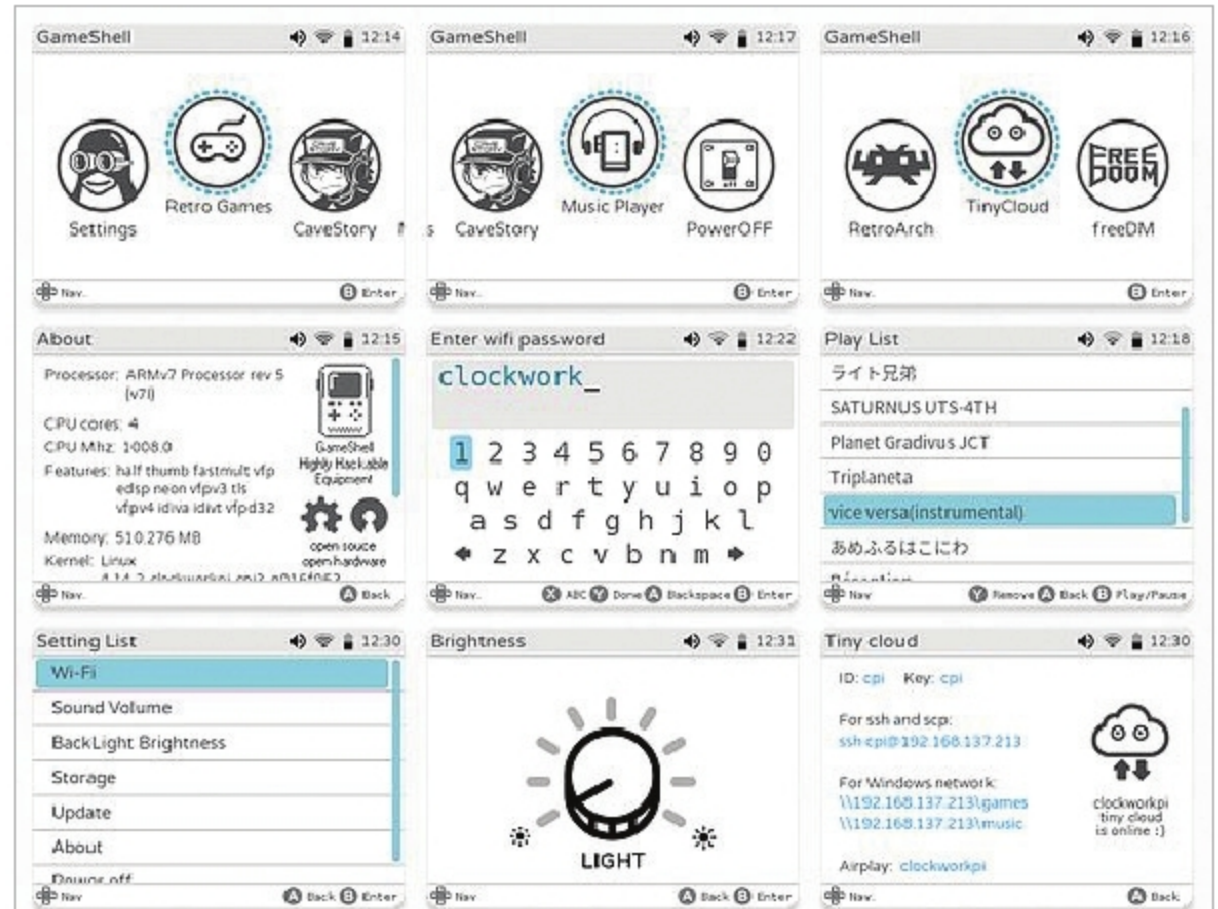


Figure 5: Clockwork Pi OS UIs

various software including PICO 8, LOVE2D, PyGame, Phaser.io, Libreto and other game engines. The operating system has a monochrome-style UI, which lets end users navigate all features of the GameShell console. The user can adjust settings, launch software, play music and even Cloudwork Pi cloud access is possible using standard controls. It includes emulators for GameBoy Advanced and the indie game Cave Story. It comes with RetroArch, acting as a frontend for emulators.

Pros

- Good for almost all age groups and easy to assemble
- Modular design that's fully open to modification
- Good build quality
- Customisation options bundled with the system
- Light bar module gives bonus buttons

Cons

- Battery limitation of 1200mAh

- Some games look squashed/cluttered with a 320 x 240 display

Final verdict

GameShell is a superb handheld gaming console and can be termed as a gamer's heaven. It is best for hard core techies and developers. It allows end users to play almost every retro game that exists till date, but requires an understanding of Linux, the command line, etc. With the updated Gamepad, GameShell is a delight to play all retro games on. The screen could have been better for an enhanced user experience.

So, if you are looking for a solid portable retro-gaming machine, GameShell is the best for you.

It is priced at US\$ 159 for users and US\$ 139 for students [<https://www.clockworkpi.com/shop-1>]. **END** 

References

- [1] <https://www.clockworkpi.com/>
- [2] <https://github.com/clockworkpi>

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